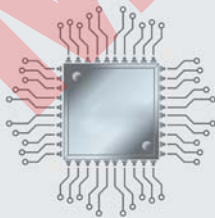


I

Digital Circuits



Contents

1. Number Systems and Binary Codes	6
2. Logic Gates	8
3. Combinational Logic Circuits	12
4. Sequential Circuits	17
5. Integrated-Circuit Logic Families	22
6. ADC and DAC	25
7. Semiconductor Memories	28

DESCRIPTION SHEET

DIGITAL CIRCUITS

Chapter-1 : Number Systems and Binary Codes

- Boolean Algebra
- Radix based number systems - Standard number systems:
 - Conversion decimal number to any radix number, any radix number to decimal number
 - $(r-1)$'s and r 's complements
 - Signed binary number representation sign magnitude, signed 1's and signal 2's complement range of number in various representation problems.
 - BCD codes: Weighted and non-weighted codes, self complementary codes
 - Code conversion:
 - ⇒ BCD to Excess-3
 - ⇒ Excess-3 to BCD
 - ⇒ Binary to gray
 - ⇒ Gray to binary

Chapter-2 : Logic Gates

- Boolean operators
- Basic logic gates
- Boolean algebra
- Sum of product and Product of sum from Truth table
- Minterms and Maxterms
- Literals and variables
- Simplification of boolean functions
- Universal logic gate
- Simplification and equality properties on EX-OR and EX-NOR
- K-Map:
 - Necessity of K-map
 - K-map on 2, 3 and 4 variables
 - Definition of implicant, prime implicant and essential prime implicant

- Finding prime implicant and essential prime implicant
- Finding minimal expressions using k-map SOP and POS
- Don't care combinations definition and problems.
- Five variable K-map

Chapter-3 : Combinational Logic Circuits

- Arithmetic Circuits:
 - Half-adder, Full-adder, Half subtractor, Full subtractor
 - Binary adder
 - ⇒ Ripple carry adder
 - ⇒ Carry look ahead adder
 - Problems with 1's and 2's complement, advantage of 2's complement
 - 4-bit binary adder
 - Binary comparator circuit
- Multiplexer:
 - MUX circuit with logic gates
 - Realizing functions using MUX
 - Analysis of MUX problems
- Decoder/DE-MUX circuits
- Encoder and priority encoders

Chapter-4 : Sequential Circuits

- Flip-Flops:
 - Binary latch and S-R and J-K flip-flop circuits with NAND gates : Truth table, characteristic table, excitation table, affect of clear and preset input
 - Race around condition and its solution
 - Master slave J-K flip-flop
 - D and T flip-flop
 - General procedure for converting one flip-flop to another flip-flop

- Registers:
 - SISO, SIPO, PISO and PIPO operations
- Asynchronous counter:
 - MOD-N counter: Ripple counter design using clear and preset terminals
- Synchronous counter:
 - Design procedure
 - Analysis of synchronous counter

Chapter-5 : Integrated-Circuit Logic Families

- DTL, TTL, NMOS and PMOS, CMOS Logic families
- Characteristic of logic IC's

Chapter-6 : ADC and DAC

- D to A converter:
 - Weighted resistor
 - R-2R and inverse R-2R
- A to D converter
 - Comparator type
 - Successive approximation type
 - Counter type

■■■■

MADE EASY

1

Number Systems and Binary Codes



Multiple Choice Questions

- Q.1** The minimum decimal equivalent of the number $11C.0$ is
 (a) 183 (b) 194
 (c) 268 (d) 269 [ESE-2000]
- Q.2** Given $(135)_{\text{base } x} + (144)_{\text{base } x} = (323)_{\text{base } x}$
 What is the value of base x ?
 (a) 5 (b) 3
 (c) 12 (d) 6 [ESE-2005]
- Q.3** If $(11X1Y)_8 = (12C9)_{16}$ then the values X and Y are
 (a) 5 and 1 (b) 5 and 7
 (c) 7 and 5 (d) 1 and 5 [ESE-2012]
- Q.4** Which of the following represents ' $E3_{16}$ '?
 (a) $(1CE)_{16} + (A2)_{16}$ (b) $(1BC)_{16} - (DE)_{16}$
 (c) $(2BC)_{16} - (1DE)_{16}$ (d) $(200)_{16} - (11D)_{16}$ [ESE-2002]
- Q.5** If $(2.3)_{\text{base } 4} + (1.2)_{\text{base } 4} = (y)_{\text{base } 4}$; what is the value of y ?
 (a) 10.1 (b) 10.01
 (c) 10.2 (d) 1.02 [ESE-2005]
- Q.6** The 2's complement representation of -17 is
 (a) 101110 (b) 101111
 (c) 111110 (d) 110001 [GATE-2001]
- Q.7** A signed integer has been stored in a byte using the 2's complement format. We wish to store the same integer in a 16 bit word. We should
 (a) copy the original byte to the less significant byte of the word and fill the more significant byte with zeros.
 (b) copy the original byte to the more significant byte of the word and fill the less significant byte with zeros.
 (c) copy the original byte to the less significant byte of the word and make each bit of the more significant byte equal to the most significant bit of the original byte.
 (d) copy the original byte to the less significant bytes well as the more significant byte of the word. [GATE-1997]
- Q.8** Sign extension is the step in
 (a) Floating point multiplication
 (b) Signed 16 bit integer addition
 (c) Arithmetic left shift
 (d) Converting a signed integer from one size to another [GATE-2002]
- Q.9** Consider the following statements:
 1. When two unsigned numbers are added, an overflow is detected from the carry into the most significant position.
 2. An overflow does not occur if the two numbers added are both negative.
 3. If the carry into the sign bit position and carry out of the sign bit position are not equal, an overflow condition is produced.
 Which of the above statement(s) is/are correct?
 (a) 1, 2 and 3 (b) 1 only
 (c) 2 only (d) 3 only [ESE-2015]
- Q.10** 11001, 1001 and 111001 correspond to the 2's complement representation of which one of the following sets of number?

© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.

- (a) 25, 9 and 57 respectively
(b) -6, -6 and -6 respectively
(c) -7, -7 and -7 respectively
(d) -25, -9 and -57 respectively

[GATE-2004]

Q.11 The range of integers that can be represented by an n bit 2's complement number system is

- (a) -2^{n-1} to $(2^{n-1} - 1)$
(b) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
(c) -2^{n-1} to 2^{n-1}
(d) $-(2^{n-1} + 1)$ to $(2^{n-1} - 1)$

[GATE-2005]

Q.12 The BCD code for a decimal number $(874)_{10}$ is

- (a) $(100001110100)_{BCD}$
(b) $(010001111000)_{BCD}$
(c) $(100001000111)_{BCD}$
(d) $(011110000100)_{BCD}$

[ESE-2013]

Q.13 A new Binary Coded Pentary (BCP) number system is proposed in which every digit of a base-5 number is represented by its corresponding 3-bit binary code. For example, the base-5 number 24 will be represented by its BCP code 010100. In this numbering system, the BCP code 100010011001 corresponds to the following number in base-5 system

- (a) 423 (b) 1324
(c) 2201 (d) 4231

[GATE-2006]

Q.14 A Gray code is a/an:

- (a) Binary weight code
(b) Arithmetic code
(c) Code which exhibits a single bit change between two successive codes
(d) Alphanumeric code

[ESE-2005]

23
69

Numerical Answer Type Questions

Q.15 Consider the number given by the decimal expression.

$$16^3 \times 9 + 16^2 \times 7 + 16 \times 5 + 3$$

The number of 1's in the unsigned binary representation of the number is

Q.16 The base (or radix) of the number system such that the following equation holds is _____.

$$\frac{312}{20} = 13.1$$

[GATE-2014]

Q.17 Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is _____.

[GATE-2014]

Q.18 The 16-bit 2's complement representation of an integer is 1111 1111 0101; its decimal representation is _____.

[GATE-2016]

Q.19 The number of bytes required to represent the decimal number 1856357 in packed BCD (Binary Coded Decimal) form is _____.

[GATE-2014]



Try Yourself

T1. A number is expressed in binary two's complement as 10011. Its decimal equivalent value is _____.

T2. If $(10)_x \times (10)_x = (100)_x$; $(100)_x \times (100)_x = (10000)_x$ then x can take value:

- (a) 2 (b) 5
(c) 10 (d) All of these

T3. Consider the following multiplication: $(10w1z)_2 \times (15)_{10} = (y01011001)_2$ Which one of the following gives appropriate values of w , y and z ?

- (a) $w = 0, y = 0, z = 1$
(b) $w = 0, y = 1, z = 1$
(c) $w = 1, y = 1, z = 1$
(d) $w = 1, y = 1, z = 0$

T4. The 9's complement of $(25.639)_{10}$ is

- (a) 74.360 (b) 0.6732
(c) 6.732 (d) 7.436

■■■■

Logic Gates



Multiple Choice Questions

- Q.1** Consider the following Sum of Products expression, F .

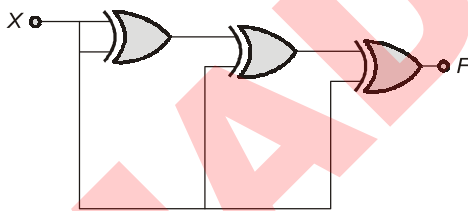
$$F = ABC + \bar{A}\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C}$$

The equivalent Product of Sums expression is

- (a) $F = (A + \bar{B} + C)(\bar{A} + B + C)(\bar{A} + \bar{B} + C)$
- (b) $F = (A + \bar{B} + \bar{C})(A + B + C)(\bar{A} + \bar{B} + \bar{C})$
- (c) $F = (\bar{A} + B + \bar{C})(A + \bar{B} + \bar{C})(A + B + C)$
- (d) $F = (\bar{A} + \bar{B} + C)(A + B + \bar{C})(A + B + C)$

[GATE-2015]

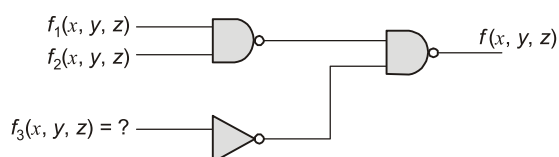
- Q.2** For the circuit shown below the output F is given by



- (a) $F = 1$
- (b) $F = 0$
- (c) $F = X$
- (d) $F = \bar{X}$

[GATE-1988]

- Q.3** Consider the following logic circuit whose inputs are functions f_1, f_2, f_3 and output is f . Given that



$$f_1(x, y, z) = \Sigma(0, 1, 3, 5),$$

$$f_2(x, y, z) = \Sigma(6, 5), \text{ and}$$

$$f(x, y, z) = \Sigma(1, 4, 5),$$

f_3 is

- (a) $\Sigma(1, 4, 5)$
- (b) $\Sigma(6, 7)$
- (c) $\Sigma(0, 1, 3, 5)$
- (d) None of these

[GATE-2002]

- Q.4** Consider the Boolean operator $\#$ with the following properties

$$x \# 0 = x, x \# 1 = \bar{x}, x \# x = 0$$

and $x \# \bar{x} = 1$. Then $x \# y$ is equivalent to

- (a) $x\bar{y} + \bar{x}y$
- (b) $x\bar{y} + \bar{x}\bar{y}$
- (c) $\bar{x}y + xy$
- (d) $xy + \bar{x}\bar{y}$

[GATE-2016]

- Q.5** Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?

- (a) $\overline{P \oplus Q} = P \odot Q$
- (b) $\bar{P} \oplus Q = P \odot Q$
- (c) $\bar{P} \oplus \bar{Q} = P \oplus Q$
- (d) $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$

[GATE-2018]

- Q.6** The minimum number of 2-input NAND gates required to implement the Boolean function $Z = \bar{A}\bar{B}C$, assuming that A, B and C are available, is

- (a) two
- (b) three
- (c) five
- (d) six

[GATE-1998]

© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.

Q.7 The minimum number of gates required to realize the function $AB + \bar{C}$ (using NAND gates only) is

- (a) 2 (b) 3
(c) 4 (d) 6

[ESE-2015]

Q.8 Minimum number of 2-input NAND gates required to implement the function,

$$F = (\bar{X} + \bar{Y})(Z + W)$$

- (a) 3 (b) 4
(c) 5 (d) 6

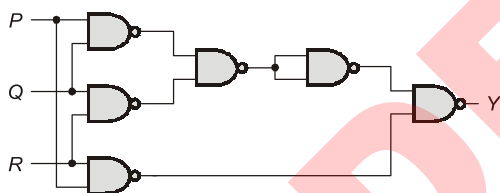
[GATE-1988]

Q.9 NAND and NOR gates are called 'Universal' gates primary because

- (a) they are available everywhere.
(b) they are widely used in I.C. packages.
(c) they can be combined to produce AND, OR and NOT gate.
(d) they can be manufactured easily.

[ESE-2016]

Q.10 The output Y in the circuit below is always "1" when



- (a) two or more of the inputs P, Q, R are "0"
(b) two or more of the inputs P, Q, R are "1"
(c) any odd number of the inputs P, Q, R is "0"
(d) any odd number of the inputs P, Q, R is "1"

[GATE-2011]

Q.11 Statement (I) : XOR gate is not a universal gate.

Statement (II) : It is not possible to realize all Boolean function using XOR gates only.

- (a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is not the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

[ESE-2016]

Q.12 Statement (I): A NAND gate represents a universal logic family.

Statement (II): Only two NAND gates are sufficient to accomplish all of the basic gates.

- (a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is not the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

[ESE-2014]

Q.13 Following is the K -map of a Boolean function of five variables P, Q, R, S and X . The minimum sum-of-product (SOP) expression for the function is

PQ \ RS	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

$X = 0$

PQ \ RS	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	0	0	0
10	0	1	1	0

$X = 1$

- (a) $\bar{P}\bar{Q}S\bar{X} + P\bar{Q}S\bar{X} + Q\bar{R}\bar{S}X + QR\bar{S}X$
(b) $\bar{Q}S\bar{X} + Q\bar{S}X$
(c) $\bar{Q}SX + Q\bar{S}\bar{X}$
(d) $\bar{Q}S + Q\bar{S}$

[GATE-2016]

Q.14 The minterm expansion of

$$f(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$$

- (a) $m_2 + m_4 + m_6 + m_7$
(b) $m_0 + m_1 + m_3 + m_5$
(c) $m_0 + m_1 + m_6 + m_1$
(d) $m_2 + m_3 + m_4 + m_5$

[GATE-2010]

Q.15 What is the minimal form of the Karnaugh map shown below? Assume that X denotes a don't care term.

cd \ ab	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

- (a) $\bar{b}\bar{d}$ (b) $\bar{b}\bar{d} + \bar{b}\bar{c}$
 (c) $\bar{b}\bar{d} + a\bar{b}\bar{c}d$ (d) $\bar{b}\bar{d} + \bar{b}\bar{c} + \bar{c}\bar{d}$

[GATE-2012]

Q.16 In the sum of products function:

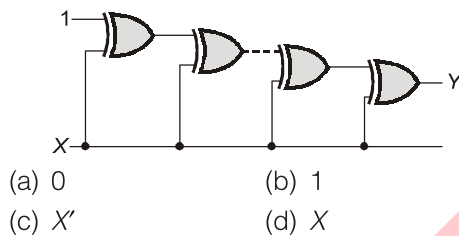
$$f(X, Y, Z) = \Sigma(2, 3, 4, 5)$$

the prime implicants are

- (a) $\bar{X}Y + X\bar{Y}$
 (b) $\bar{X}Y + X\bar{Y}\bar{Z} + X\bar{Y}Z$
 (c) $\bar{X}Y\bar{Z} + \bar{X}YZ + X\bar{Y}$
 (d) $\bar{X}Y\bar{Z} + \bar{X}YZ + X\bar{Y}\bar{Z} + X\bar{Y}Z$

[GATE-2012]

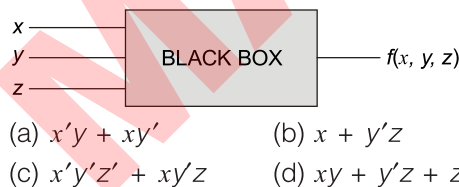
Q.17 If the input to the digital circuit of the below figure consisting of a cascade of 20 XOR gates is X , then what is the output Y ?



[ESE-2007]

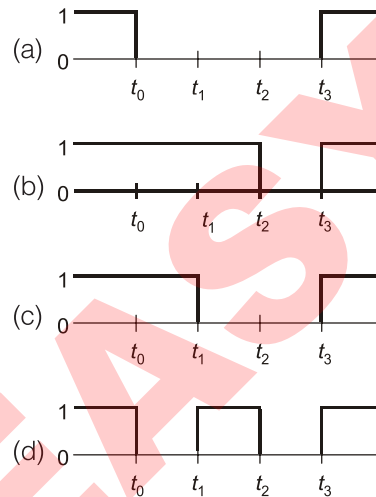
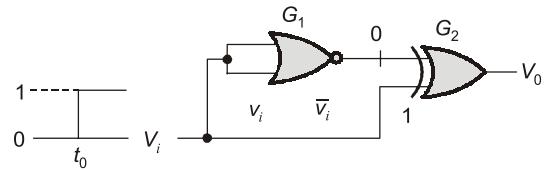
Q.18 The black box in the figure shown below consists of a minimum complexity circuit that uses only AND, OR and NOT gates.

The function $f(x, y, z) = 1$ whenever x, y are different and 0 otherwise. In addition the 3 inputs x, y, z are never all the same value. Which one of the following equations leads to the correct design for the minimum complexity circuit?



[ESE-2007]

Q.19 The gates G_1 and G_2 in the figure have propagation delays of 10 nsec and 20 nsec respectively. If the input V_i makes an abrupt change from logic 0 to 1 at time $t = t_0$, then the output waveform V_0 is



[GATE-2002]

Q.20 Let $f(x, y, z) = \bar{x} + \bar{y}z + xz$ be a switching function. Which one of the following is valid?

- (a) $\bar{y}x$ is a prime implicant of f
 (b) xz is a minterm of f
 (c) xz is an implicant of f
 (d) y is a prime implicant of f

[GATE-1997]

Q.21 For the identity $AB + \bar{A}C + BC = AB + \bar{A}C$, the dual form is

- (a) $(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$
 (b) $(\bar{A} + \bar{B})(A + \bar{C})(\bar{B} + \bar{C}) = (\bar{A} + \bar{B})(A + \bar{C})$
 (c) $(A + B)(\bar{A} + C)(B + C) = (\bar{A} + \bar{B})(A + C)$
 (d) $\bar{A}\bar{B} + A\bar{C} + \bar{B}\bar{C} = \bar{A}\bar{B} + A\bar{C}$

[GATE-1998]

Q.22 The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot, ',)$, written as F^D , is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

- (a) 2^n (b) 2^{n-1}
 (c) 2^{2^n} (d) $2^{2^{n-1}}$

[GATE-2014]

23
69

Numerical Answer Type Questions

Q.23 Consider the Karnaugh map given below, where X represents “don’t care” and blank represents 0.

$\begin{matrix} ba \\ dc \end{matrix}$	00	01	11	10
00		X	X	
01	1			X
11	1			1
10		X	X	

Assume for all inputs (a, b, c, d), the respective complements ($\bar{a}, \bar{b}, \bar{c}, \bar{d}$) are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

[GATE-2017]

Q.24 The number of essential prime-implicants in the given function $f(w, x, y, z) = \sum m(0, 2, 6, 7, 8, 9, 13, 15)$ is _____.

Q.25 Consider the minterm list form of a Boolean function F given below:

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

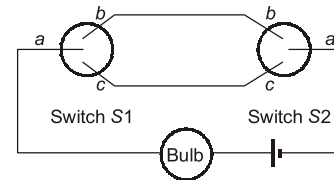
Here, m denotes a minterm and d denotes a don’t care term. The number of essential prime implicants of the function F is _____.

[GATE-2018]



Try Yourself

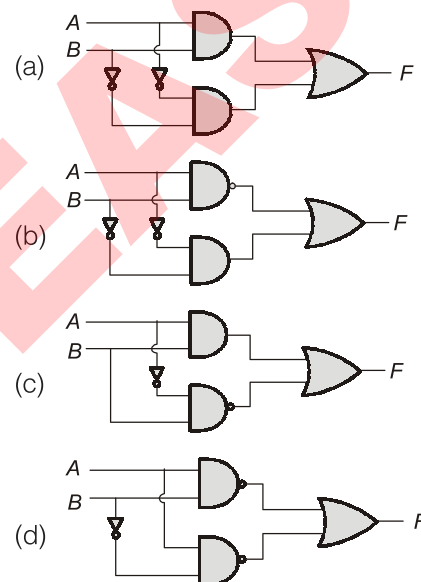
T1. A two-way switch has three terminals a, b and c . In ON position (logic value 1), a is connected to b , and in OFF position, a is connected to c . Two of these two-way switches $S1$ and $S2$ are connected to a bulb as shown below.



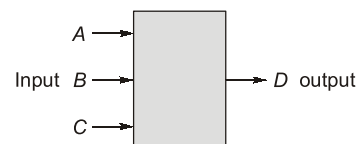
Which of the following expressions, if true, will always result in the lighting of the bulb?

- (a) $S1 \cdot \bar{S2}$ (b) $S1 + S2$
(c) $\bar{S1} \oplus \bar{S2}$ (d) $S1 \oplus S2$

T2. Which one of the following figures represents the coincidence logic?



T3. For the box shown the output D is true if and only if a majority of the inputs are true.



The Boolean function for the output is

- (a) $D = AB\bar{C} + \bar{A}BC + A\bar{B}C$
(b) $D = ABC + \bar{A}BC + A\bar{B}C + AB\bar{C}$
(c) $D = \bar{A}\bar{B}\bar{C} + AB + AC + BC$
(d) $D = \bar{A}\bar{B}C + A\bar{B}\bar{C} + \bar{A}B\bar{C} + ABC$



Combinational Logic Circuits



Multiple Choice Questions

Q.1 The following switching functions are to be implemented using a Decoder:

$$f_1 = \sum m(1, 2, 4, 8, 10, 14)$$

$$f_2 = \sum m(2, 5, 9, 11)$$

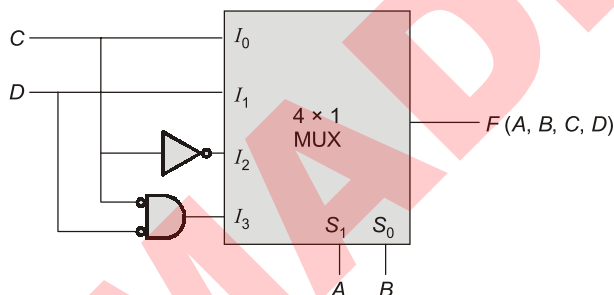
$$f_3 = \sum m(2, 4, 5, 6, 7)$$

The minimum configuration of the decoder should be

- (a) 2-to-4 lines (b) 3-to-8 lines
(c) 4-to-16 lines (d) 5-to-32 lines

[ESE-2015]

Q.2 The Boolean function realized by the logic circuit shown is



- (a) $F = \sum m(0, 1, 3, 5, 9, 10, 14)$
(b) $F = \sum m(2, 3, 5, 7, 8, 12, 13)$
(c) $F = \sum m(1, 2, 4, 5, 11, 14, 15)$
(d) $F = \sum m(2, 3, 5, 7, 8, 9, 12)$

[GATE-2010]

Q.3 Consider the following statements:

A multiplexer

1. selects one of the several inputs and transmits it to a single output
2. routes the data from a single input to one of many output

3. converts parallel data into serial data

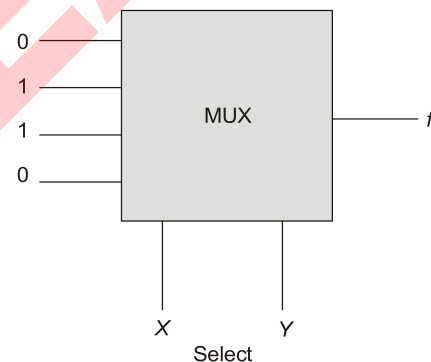
4. is a combinational circuit

Which of these statements are correct?

- (a) 1, 2 and 4 (b) 2, 3, and 4
(c) 1, 3 and 4 (d) 1, 2 and 3

[ESE-2000]

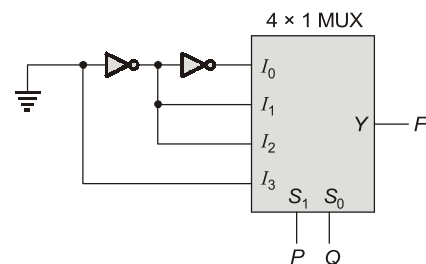
Q.4 What is the output $f(x, y)$ of the multiplexer resulting from the input logical values?



- (a) An EXOR gate (b) A NOR gate
(c) An AND gate (d) A NAND gate

[ESE-2005]

Q.5 The logic function implemented by the circuit below is (ground implies a logic "0")

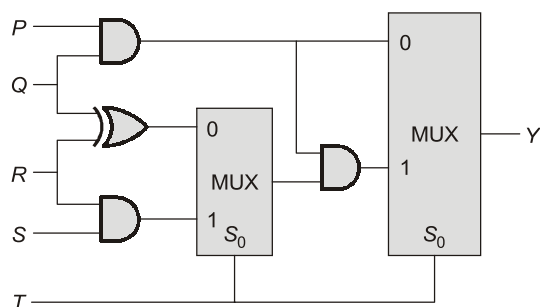


- (a) $F = \text{AND}(P, Q)$ (b) $F = \text{OR}(P, Q)$
(c) $F = \text{XNOR}(P, Q)$ (d) $F = \text{XOR}(P, Q)$

[GATE-2011]

© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.

- Q.6** The propagation delays of the XOR gate, AND gate and multiplexer (MUX) in the circuit shown in the figure are 4 ns, 2 ns and 1 ns, respectively.



If all the inputs P , Q , R , S and T are applied simultaneously and held constant, the maximum propagation delay of the circuit is

- (a) 3 ns (b) 5 ns
(c) 6 ns (d) 7 ns

[2021 : 2 Marks]

- Q.7** Without any additional circuitry, an 8 :1 MUX can be used to obtain

- (a) some but not all Boolean functions of 3 variables
(b) all functions of 3 variables but none of 4 variables
(c) all functions of 3 variables and some but not all of 4 variables
(d) all functions of 4 variables

[GATE-2003]

- Q.8** The addition of two binary variables A and B results into a SUM and a CARRY output. Consider the following expressions for the SUM and CARRY outputs.

1. $SUM = A \cdot B + \bar{A} \cdot \bar{B}$
2. $SUM = A \cdot \bar{B} + \bar{A} \cdot B$
3. $CARRY = A \cdot B$
4. $CARRY = A + B$

Which of these expressions are correct?

- (a) 1 and 3 (b) 2 and 3
(c) 2 and 4 (d) 1 and 4

[ESE-2003]

- Q.9** Consider the following statements:
For 3 input variables a , b , c ; a Boolean function $y = ab + bc + ca$ represents

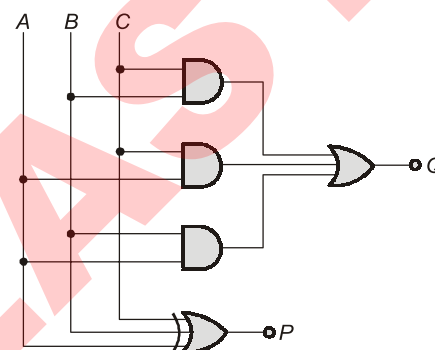
1. a 3-input majority gate
2. a 3-input minority gate
3. carry output of a full adder
4. product circuit for a , b and c

Which of the above statements are correct?

- (a) 1 and 4 only (b) 2 and 3 only
(c) 1 and 3 only (d) 3 and 4 only

[ESE-2007]

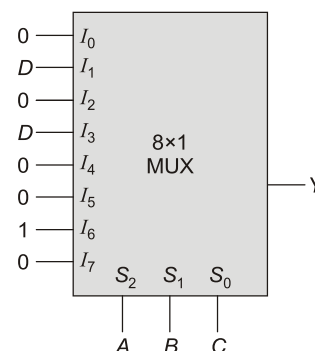
- Q.10** In the circuit of figure A , B , C are the inputs and P , Q are the two outputs. The circuit is a



- (a) half adder where P is the sum and Q is the carry
(b) half adder where P is the carry and Q is the sum
(c) full adder where P is the sum and Q is the carry
(d) full adder where P is the carry and Q is the sum

[GATE-2004]

- Q.11** An 8-to-1 multiplexer is used to implement a logical function Y as shown in the figure. The output



- (a) $Y = \bar{A}\bar{B}C + A\bar{C}D$
 (b) $Y = \bar{A}BC + A\bar{B}D$
 (c) $Y = AB\bar{C} + \bar{A}CD$
 (d) $Y = \bar{A}\bar{B}D + A\bar{B}C$

[2014 : 2 Marks, Set-4]

Q.12 For an n -bit binary adder, what is the number of gates through which a carry has to propagate from input to output?

- (a) n (b) $2n$
 (c) n^2 (d) $n + 1$

[ESE-2016]

Q.13 How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

- (a) 7 (b) 8
 (c) 9 (d) 10

[GATE-2007]

Q.14 For a binary half-subtractor having two inputs A and B , the correct sets of logical expressions for the output D ($= A$ minus B) and X ($=$ borrow) are

- (a) $D = AB + \bar{A}B, X = \bar{A}B$
 (b) $D = \bar{A}B + A\bar{B}, X = A\bar{B}$
 (c) $D = \bar{A}B + A\bar{B}, X = \bar{A}B$
 (d) $D = AB + \bar{A}\bar{B}, X = A\bar{B}$

[ESE-2003]

Q.15 When two 8-bit numbers $A_7 \dots A_0$ and $B_7 \dots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a ripple-carry adder, the sum bits obtained are $S_7 \dots S_0$ and the carry bits are $C_7 \dots C_0$. An overflow is said to have occurred if

- (a) the carry bit C_7 is 1
 (b) all the carry bits ($C_7, \dots, C_0$) are 1
 (c) $(A_7 \cdot B_7 \cdot \bar{S}_7 + \bar{A}_7 \cdot \bar{B}_7 \cdot S_7)$ is 1
 (d) $(A_0 \cdot B_0 \cdot \bar{S}_0 + \bar{A}_0 \cdot \bar{B}_0 \cdot S_0)$ is 1

[GATE-2017]

23
69

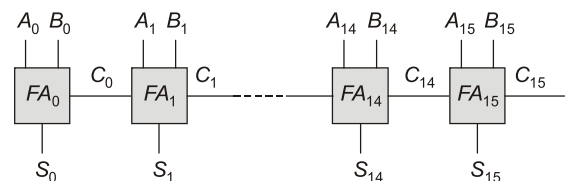
Numerical Answer Type Questions

Q.16 A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds.

A 4-bit ripple-carry binary adder is implemented by using four full adders. The total propagation time of this 4-bit binary adder in microseconds is _____.

Q.17 A one bit full adder takes 75 nsec to produce sum and 50 nsec to produce carry. A 4 bit parallel adder is designed using this type of full adder. The maximum rate of additions per second can be provided by 4 bit parallel adder is $A \times 10^6$ additions/sec. The value of A is _____.

Q.18 A 16-bit ripple carry adder is realized using 16 identical full adders (FA) as shown in the figure. The carry-propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of this 16-bit adder will be _____.

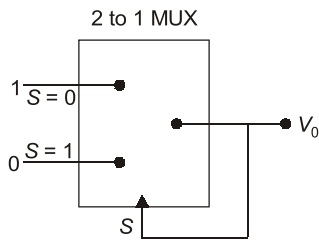


Q.19 Minimum number of NAND gates required to implement Sum in half-adder circuit is _____.



Try Yourself

T1. A 2-to-1 digital multiplexer having a switching delay of $1 \mu s$ is connected as shown in the figure. The output of the multiplexer is tied to its own select input S . The input which gets selected when $S = 0$ is tied to 1 and the input that gets selected when $S = 1$ is tied to 0. The output V_0 will be



- (a) 0
(b) 1
(c) Pulse train of frequency 0.5 MHz
(d) Pulse train of frequency 1.0 MHz

T2. Consider the multiplexer with X and Y as data inputs and Z as control input. $Z = 0$ selects input X and $Z = 1$ selects input Y . What are the connections required to realize the 2-variable Boolean function $f = T + R$, without using any additional hardware?

- (a) R to X , 1 to Y , T to Z
(b) T to X , R to Y , T to Z
(c) T to X , R to Y , 0 to Z
(d) R to X , 0 to Y , T to Z

T3. The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B . The number of combinations for which the output is logic 1, is

- (a) 4 (b) 6
(c) 8 (d) 10

T4. Consider two 4-bit numbers $A = A_3 A_2 A_1 A_0$ and $B = B_3 B_2 B_1 B_0$ and the expression $x_i = A_i B_i + \overline{A_i} \overline{B_i}$ for $i = 0, 1, 2, 3$. The expression

$A_3 \overline{B_3} + x_3 A_2 \overline{B_2} + x_3 x_2 A_1 \overline{B_1} + x_3 x_2 x_1 A_0 \overline{B_0}$
evaluates to 1 if

- (a) $A = B$ (b) $A \neq B$
(c) $A > B$ (d) $A < B$

T5. In a look-ahead carry generator, the carry generate function G_i and the carry propagate function P_i for inputs, A_i and B_i are given by

$$P_i = A_i \oplus B_i \text{ and } G_i = A_i B_i$$

The expressions for the sum bit S_i and carry bit C_{i+1} of the look-ahead carry adder are given by

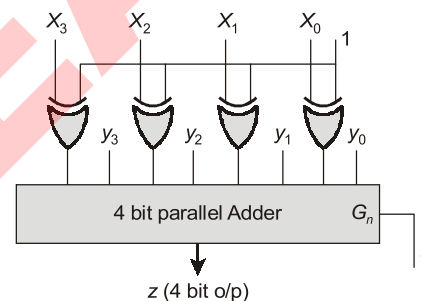
© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.

$S_i = P_i \oplus C_i$ and $C_{i+1} = G_i + P_i C_i$, where C_0 is the input carry.

Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i and G_i are available for the carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit adder with S_3, S_2, S_1, S_0 and C_4 as its outputs are respectively

- (a) 6, 3 (b) 10, 4
(c) 6, 4 (d) 10, 5

T6. Identify the correct statement with respect to following circuit? Numbers are represented in signed magnitude format.



- (a) It outputs $x + y$
(b) It outputs $y - x$
(c) It outputs $x + 1$
(d) It outputs $y + 1$

Statement for Linked Answer Question T7 and T8.

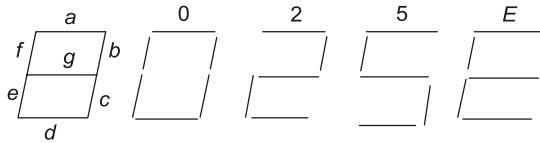
Two products are sold from a vending machine, which has two push buttons P_1 and P_2 . When a button is pressed, the price of the corresponding product is displayed in a 7-segment display. If no buttons are pressed, '0' is displayed, signifying 'Rs. 0'

If only P_1 is pressed, '2' is displayed, signifying 'Rs. 2'

If only P_2 is pressed, '5' is displayed, signifying 'Rs. 5'

If both P_1 and P_2 are pressed, 'E' is displayed, signifying 'Error'

The names of the segments in the 7-segment display, and the glow of the display for '0', '2', '5' and 'E' are shown below.



Consider

- (i) push button pressed/not pressed in equivalent to logic 1/0 respectively,
- (ii) a segment glowing/not glowing in the display is equivalent to logic 1/0 respectively

T7. If segments a to g are considered as functions of P_1 and P_2 , then which of the following is correct?

(a) $g = \bar{P}_1 + P_2, d = c + e$

(b) $g = P_1 + P_2, d = c + e$

(c) $g = \bar{P}_1 + P_2, e = b + c$

(d) $g = P_1 + P_2, e = b + c$

[2009 : 2 Marks]

T8. What are the minimum numbers of NOT gates and 2-input OR gates required to design the logic of the driver for this 7-segment display?

(a) 3 NOT and 4 OR (b) 2 NOT and 4 OR

(c) 1 NOT and 3 OR (d) 2 NOT and 3 OR

[2009 : 2 Marks]

■■■■

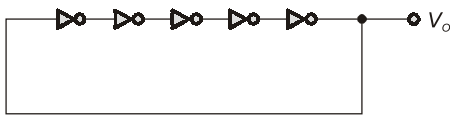
4

Sequential Circuits



Multiple Choice Questions

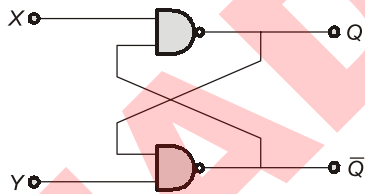
Q.1 For the ring oscillator shown in the figure, the propagation delay of each inverter is 100 pico sec. What is the fundamental frequency of the oscillator output?



- (a) 10 MHz (b) 100 MHz
(c) 1 GHz (d) 2 GHz

[GATE-2001]

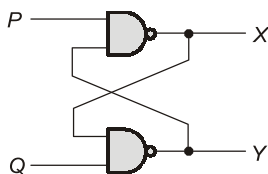
Q.2 For a flip-flop formed two NAND gates as shown in figure. The unusable state corresponds to



- (a) $X = 0, Y = 0$ (b) $X = 0, Y = 1$
(c) $X = 1, Y = 0$ (d) $X = 1, Y = \infty$

[GATE-1999]

Q.3 In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delays. The present input condition is: $P = Q = '0'$. If the input condition is changed simultaneously to $P = Q = '1'$, the outputs X and Y are



- (a) $X = '1', Y = '1'$
(b) either $X = '1', Y = '0'$ or $X = '0', Y = '1'$
(c) either $X = '1', Y = '1'$ or $X = '0', Y = '0'$
(d) $X = '0', Y = '0'$ [GATE-2017]

Q.4 An X - Y flip flop, whose Characteristic Table is given below is to be implemented using a J - K flip flop.

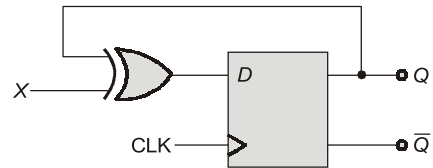
X	Y	Q_{n+1}
0	0	1
0	1	Q_n
1	0	$\bar{Q}_n$
1	1	0

This can be done by making

- (a) $J = X, K = \bar{Y}$ (b) $J = \bar{X}, K = Y$
(c) $J = Y, K = \bar{X}$ (d) $J = \bar{Y}, K = X$

[GATE-2003]

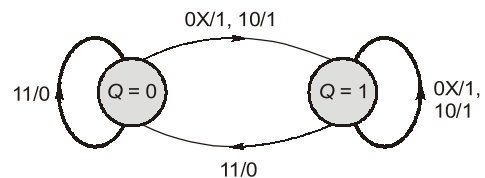
Q.5 The digital circuit shown in the figure works as



- (a) JK flip-flop (b) Clocked RS flip-flop
(c) T flip-flop (d) Ring counter

[GATE-2005]

Q.6 A state diagram of a logic which exhibits a delay in the output is shown in the figure, where X is the do not care condition, and Q is the output representing the state.



© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.

The logic gate represented by the state diagram is

- (a) XOR (b) OR
(c) AND (d) NAND

[GATE-2014]

Q.7 A cascade of three identical modulo-5 counters has an overall modulus of

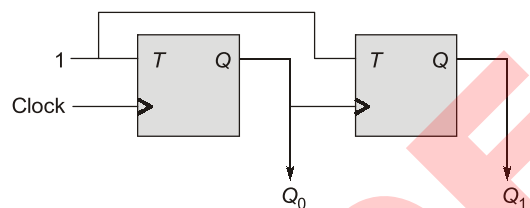
- (a) 5 (b) 25
(c) 125 (d) 625 [GATE-2014]

Q.8 A 3-bit ripple counter is constructed using three T flip-flops to do the binary counting. The three flip-flops have T-inputs fixed at

- (a) 0, 0 and 1 (b) 1, 0 and 1
(c) 0, 1 and 1 (d) 1, 1 and 1

[ESE-2016]

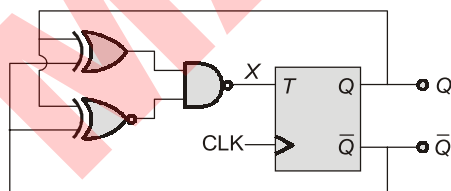
Q.9 In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00, what are the next four values of Q_1Q_0 ?



- (a) 11,10,01,00 (b) 10,11,01,00
(c) 10,00,01,11 (d) 11,10,00,01

[GATE-2010]

Q.10 The clock frequency applied to the digital circuit shown in figure below is 1 kHz. If the initial state of the output Q of the flip-flop is '0', then the frequency of the output waveform Q in kHz is



- (a) 0.25 (b) 0.5
(c) 1 (d) 2 [GATE-2013]

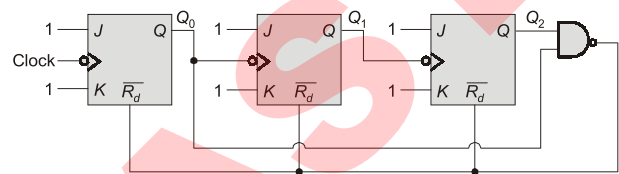
Q.11 A 4-bit modulo-16 ripple counter uses JK flip-flops. If the propagation delay of each FF is

50 ns, the maximum clock frequency that can be used is equal to:

- (a) 20 MHz (b) 10 MHz
(c) 5 MHz (d) 4 MHz

[GATE-1990]

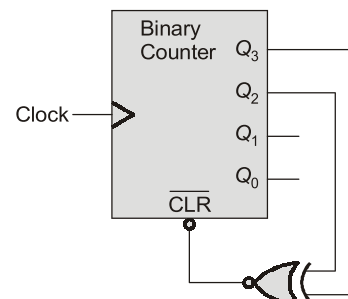
Q.12 The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset ($\overline{R_d}$ input). The counter corresponding to this circuit is



- (a) a modulo-5 binary up counter
(b) a modulo-6 binary down counter
(c) a modulo-5 binary down counter
(d) a modulo-6 binary up counter

[GATE-2015]

Q.13 The figure shows a binary counter with synchronous clear input. With the decoding logic shown, the counter works as a



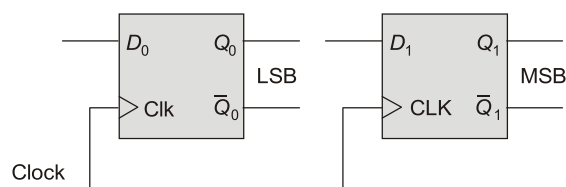
- (a) mod-2 counter (b) mod-4 counter
(c) mod-5 counter (d) mod-6 counter

[GATE-2015]

Q.14 Two D flip-flops, as shown below, are to be connected as a synchronous counter that goes through the following Q_1Q_0 sequence

00 $\rightarrow$ 01 $\rightarrow$ 11 $\rightarrow$ 10 $\rightarrow$ 00 $\rightarrow$

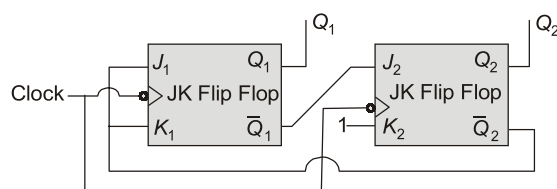
The inputs D_0 and D_1 respectively should be connected as



- (a) $\bar{Q}_1$ and Q_0 (b) $\bar{Q}_0$ and Q_1
(c) $\bar{Q}_1 Q_0$ and $\bar{Q}_1 \bar{Q}_0$ (d) $\bar{Q}_1 \bar{Q}_0$ and $Q_1 Q_0$

[GATE-2006]

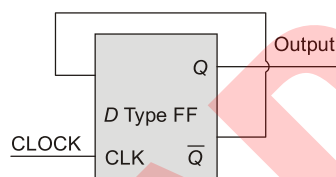
Q.15 What are the counting states (Q_1, Q_2) for the counter shown in the figure below?



- (a) 11, 10, 00, 11, 10, ...
(b) 01, 10, 11, 00, 01, ...
(c) 00, 11, 01, 10, 00, ...
(d) 01, 10, 00, 01, 10, ...

[GATE-2009]

Q.16 What is the frequency of the output Q for the circuit shown in the figure?



- (a) Twice the input clock frequency
(b) Half the input clock frequency
(c) Same as the input clock frequency
(d) Inverse of the propagation delay of the flip-flop

[ESE-2015]

Q.17 A sequence detector is designed to detect precisely 3 digital inputs, with overlapping sequences detectable. For the sequence (1, 0, 1) and input data (1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0): what is the output of this detector?

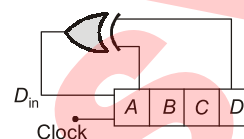
- (a) 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0
(b) 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0
(c) 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0
(d) 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0

[GATE-2020]

23
69

Numerical Answer Type Questions

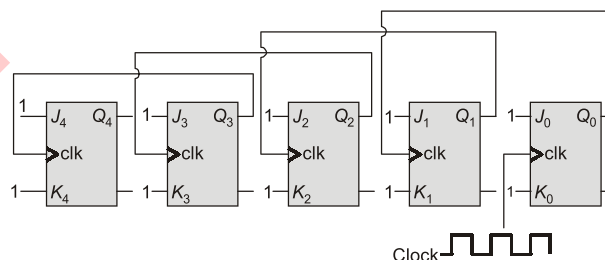
Q.18 A 4-bit shift register circuit configured for right-shift operation, i.e. $D_{in} \rightarrow A, A \rightarrow B, B \rightarrow C, C \rightarrow D$, is shown. If the present state of the shift register is $ABCD = 1101$, the number of clock cycles required to reach the state $ABCD = 1111$ is ____.



[GATE-2017]

Q.19 The minimum number of flip-flops required by a module-8 counter is ____.

Q.20 Five JK flip-flops are cascaded to form the circuit shown in Figure. Clock pulses at a frequency of 1 MHz are applied as shown. The frequency (in kHz) of the waveform at Q_3 is ____.

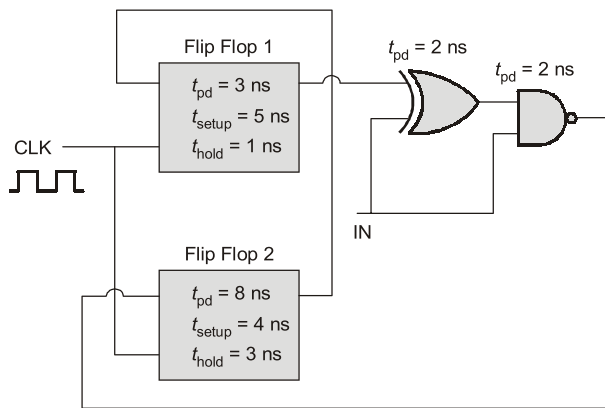


[GATE-2014]



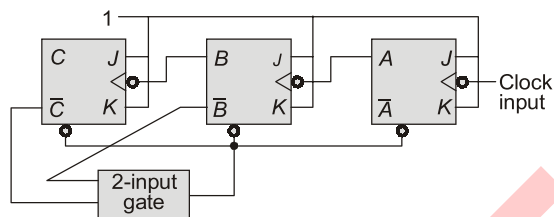
Try Yourself

T1. For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency (rounded off to the nearest integer), at which the given circuit can operate reliably, is ____ MHz.



[GATE-2020]

- T2. In the modulo-6 ripple counter shown in the figure, the output of the 2-input gate is used to clear the J - K flip-flops.

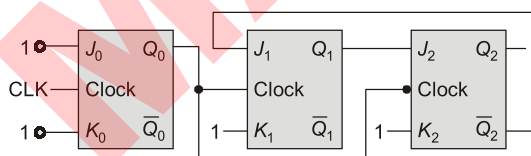


The 2-input gate is

- (a) a NAND gate (b) a NOR gate
(c) an OR gate (d) an AND gate

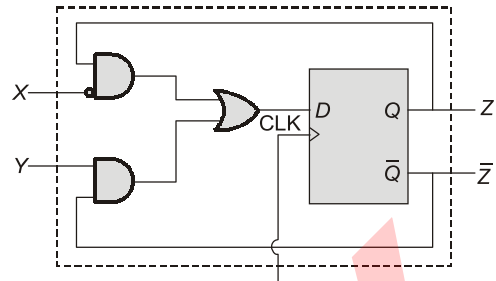
[GATE-2004]

- T3. The figure shows a digital circuit constructed using negative edge triggered J - K flip flops. Assume a starting state of $Q_2Q_1Q_0 = 000$ will repeat after _____ number of cycles of the clock CLK



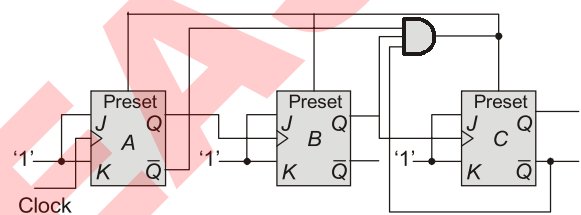
[GATE-2015]

- T4. A sequential circuit using D Flip-Flop and logic gates is shown in the figure, where X and Y are the inputs and Z is the output. The circuit is



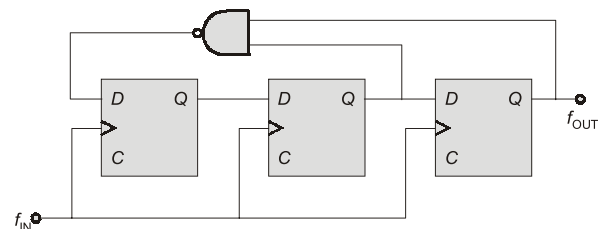
- (a) S - R Flip-Flop with inputs $X = R$ and $Y = S$
(b) S - R Flip-Flop with inputs $X = S$ and $Y = R$
(c) J - K Flip-Flop with inputs $X = J$ and $Y = K$
(d) J - K Flip-Flop with inputs $X = K$ and $Y = J$

- T5. The ripple counter shown in the figure works as a



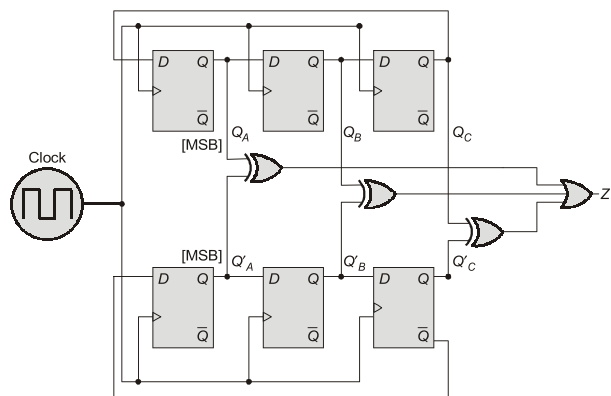
- (a) mod-3 up counter
(b) mod-5 up counter
(c) mod-3 down counter
(d) mod-5 down counter

- T6. Which one of the following statements is true about the digital circuit shown in the figure?



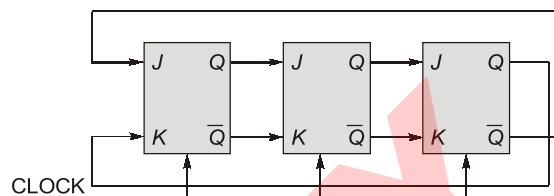
- (a) It can be used for dividing the input frequency by 3.
(b) It can be used for dividing input frequency by 5.
(c) It can be used for dividing the input frequency by 7.
(d) It cannot be reliably used as a frequency divider due to disjoint internal cycles.

- T7. For the synchronous sequential circuit shown below, the output Z is zero for the initial conditions $Q_A Q_B Q_C = Q'_A Q'_B Q'_C = 100$.



The minimum number of clock cycles after which the output z would again become zero is _____.

- T8. For the initial state of 000, the function performed by the arrangement of the J - K flip flops in the figure



- (a) Shift Register (b) Mod-3 counter
(c) Mod-6 counter (d) Mod-2 counter
- T9. How many pulses are needed to change the contents of a 8-bit upcounter from 10101100 to 00100111 (rightmost bit is the LSB)?
- (a) 134 (b) 133
(c) 124 (d) 123

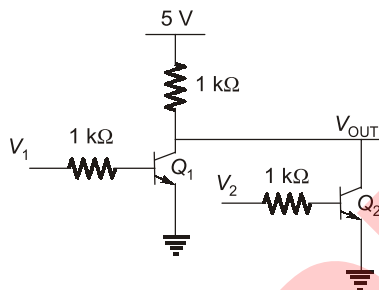
■■■■

Integrated-Circuit Logic Families



Multiple Choice Questions

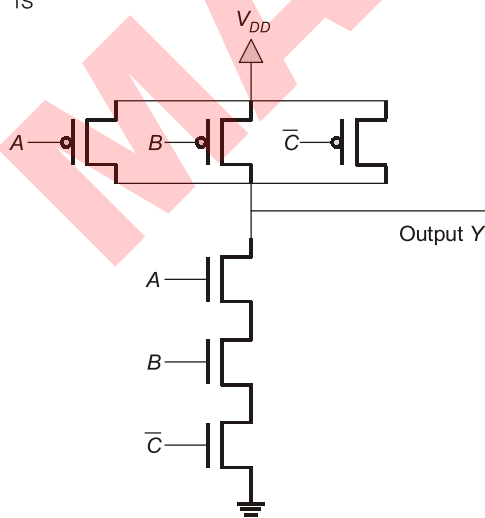
Q.1 The logical gate implemented using the circuit shown below where, V_1 and V_2 are inputs (with 0 V as digital 0 and 5 V as digital 1) and V_{OUT} is the output, is



- (a) NOT (b) NOR
(c) NAND (d) XOR

[GATE-2017]

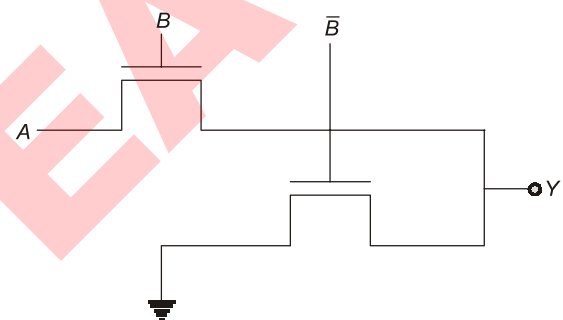
Q.2 The output (Y) of the circuit shown in the figure is



- (a) $\bar{A} + \bar{B} + C$ (b) $A + \bar{B} \cdot \bar{C} + A \cdot \bar{C}$
(c) $\bar{A} + B + \bar{C}$ (d) $A \cdot B \cdot \bar{C}$

[GATE-2014]

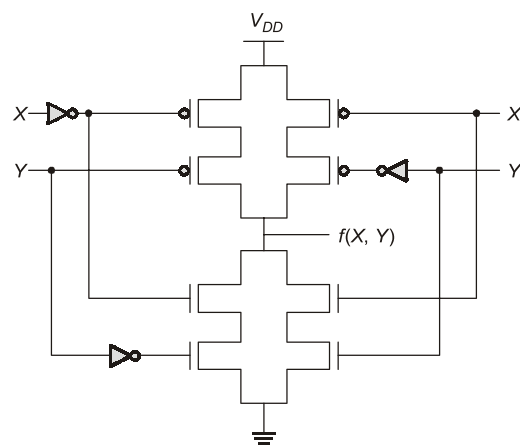
Q.3 The logic functionality realized by the circuit shown below is



- (a) OR (b) XOR
(c) NAND (d) AND

[GATE-2016]

Q.4 The logic $f(X, Y)$ realized by the given circuit is



- (a) NOR (b) AND
(c) NAND (d) XOR

[GATE-2018]

© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.

23
69

Numerical Answer Type Questions

Q.5 The inverter 74 AL S01 has the following specifications:

$$I_{OH\max} = -0.4 \text{ mA}, I_{OL\max} = 8 \text{ mA},$$

$$I_{IH\max} = 20 \text{ } \mu\text{A}, I_{IL\max} = -0.1 \text{ mA}.$$

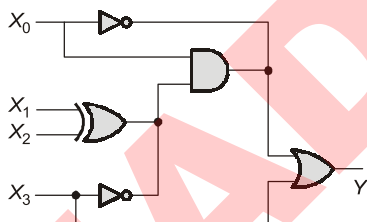
The fan out based on the above will be_____.

Q.6 An IC family has an average propagation delay of 10 ns and an average power dissipation of 5 mW. Figure of merit of IC family is_____ pJ.



Try Yourself

T1. The logic gates shown in the digital circuit below use strong pull-down nMOS transistors for LOW logic level at the outputs. When the pull-downs are off, high-value resistors set the output logic levels to HIGH (i.e. the pull-ups are weak). Note that some nodes are intentionally shorted to implement "wired logic". Such shorted nodes will be HIGH only if the outputs of all the gates whose outputs are shorted are HIGH.

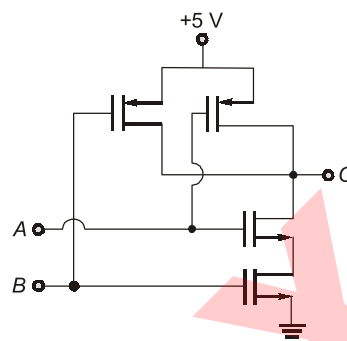


The number of distinct values of $X_3 X_2 X_1 X_0$ (out of the 16 possible values) that give $Y = 1$ is _____.

[GATE-2018]

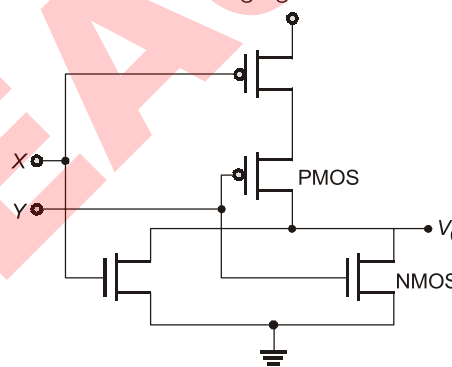
T2. In the negative logic system,
(a) the more negative of the two logic levels represents a logic '1' state
(b) the more negative of the two logic levels represents a logic '0' state
(c) all input and output voltage levels are negative
(d) the output is always complement of the intended logic function

T3. Identify the logic gate given in the figure.



- (a) NOR (b) NAND
(c) AND (d) OR

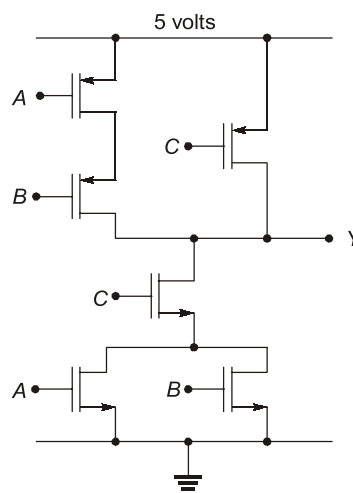
T4. A CMOS implementation of a logic gate is shown in the following figure:



The Boolean logic function realized by the circuit is

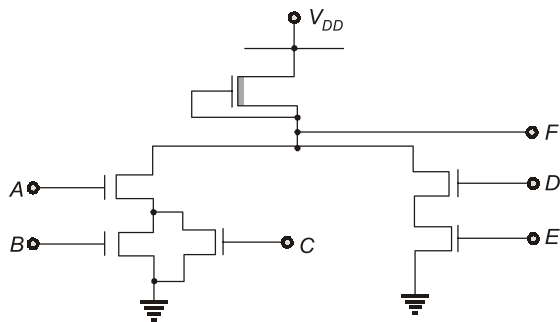
- (a) AND (b) NAND
(c) NOR (d) OR

T5. In the circuit shown



- (a) $Y = \bar{A}\bar{B} + \bar{C}$ (b) $Y = (A + B)C$
(c) $Y = (\bar{A} + \bar{B})\bar{C}$ (d) $Y = AB + C$

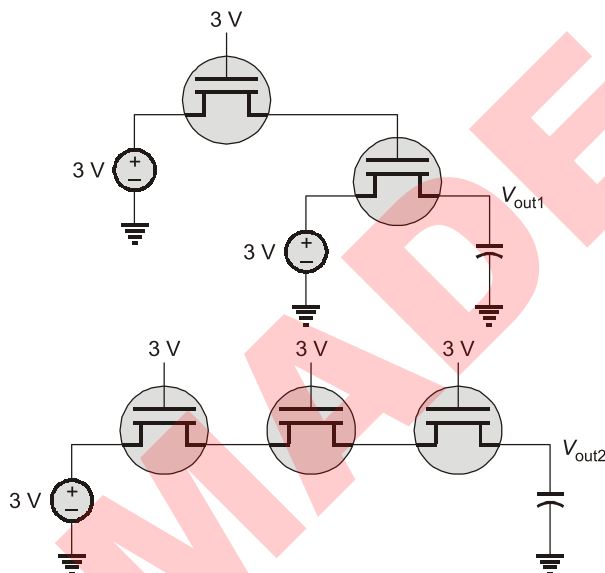
- T6. For the NMOS logic gate shown in figure, the logic function implemented is



- (a) $\overline{ABCDE}$ (b) $(AB + \overline{C}) \cdot (\overline{D} + \overline{E})$
(c) $\overline{A \cdot (B + C) + D \cdot E}$ (d) $\overline{(A + B) \cdot C + \overline{D} \cdot \overline{E}}$

[GATE-1997]

- T7. In the circuits shown, the threshold voltage of each nMOS transistor is 0.6 V. Ignoring the effect of channel length modulation and body bias, the values of V_{out1} and V_{out2} , respectively, in volts, are



- (a) 1.8 and 2.4 (b) 2.4 and 1.2
(c) 1.8 and 1.2 (d) 2.4 and 2.4

[GATE-2019]

- T8. Match List-I with List-II and select the correct answer using the code given below the Lists:

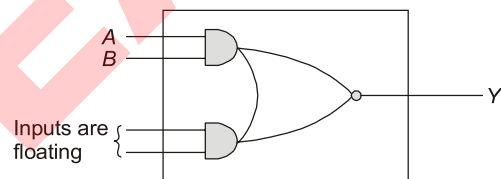
List-I List-II

- | | |
|-----------|------------------------------------|
| A. HTL | 1. High fan-out |
| B. CMOS | 2. Highest speed of operation |
| C. I^2L | 3. High noise immunity |
| D. ECL | 4. Lowest product of power & delay |

Codes:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 3 | 4 | 1 | 2 |
| (b) | 2 | 4 | 1 | 3 |
| (c) | 3 | 1 | 4 | 2 |
| (d) | 2 | 1 | 4 | 3 |

- T9. The figure shows the internal schematic of a TTL AND-OR-Invert (AOI) gate. For the inputs shown in the figure, the output Y is



- (a) 0 (b) 1
(c) AB (d) $\overline{AB}$

- T10. Among the digital IC-families-ECL, TTL and CMOS:

- (a) ECL has the least propagation delay
(b) TTL has the largest fan-out
(c) CMOS has the biggest noise margin
(d) TTL has the lowest power consumption

■■■■

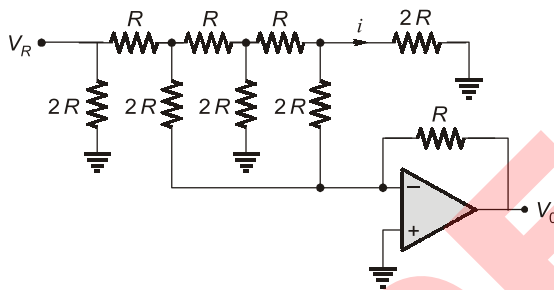
ADC and DAC



Multiple Choice Questions

Statement for Linked Answer Questions (1 and 2):

In the Digital-to-Analog converter circuit shown in the figure below, $V_R = 10\text{ V}$ and $R = 10\text{ k}\Omega$.



Q.1 The current i is

- (a) $31.25\text{ }\mu\text{A}$ (b) $62.5\text{ }\mu\text{A}$
(c) $125\text{ }\mu\text{A}$ (d) $250\text{ }\mu\text{A}$

[GATE-2007]

Q.2 The voltage V_0 is

- (a) -0.781 V (b) -1.562 V
(c) -3.125 V (d) -6.250 V

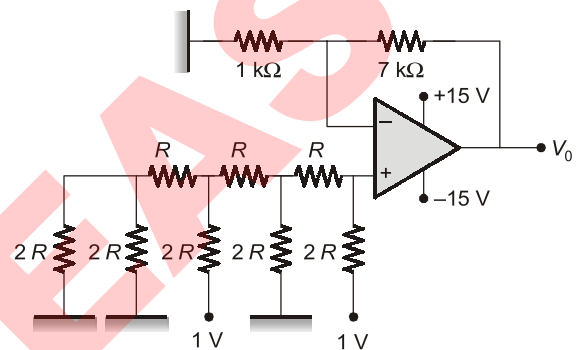
[GATE-2007]

Q.3 An 8 bit successive approximation analog to digital converter has full scale reading of 2.55 V and its conversion time for an analog input of 1 V is $20\text{ }\mu\text{s}$. The conversion time for a 2 V input will be

- (a) $10\text{ }\mu\text{s}$ (b) $20\text{ }\mu\text{s}$
(c) $40\text{ }\mu\text{s}$ (d) $50\text{ }\mu\text{s}$

[GATE-2000]

Q.4 For the 4 bit DAC shown in the figure, the output voltage V_0 is

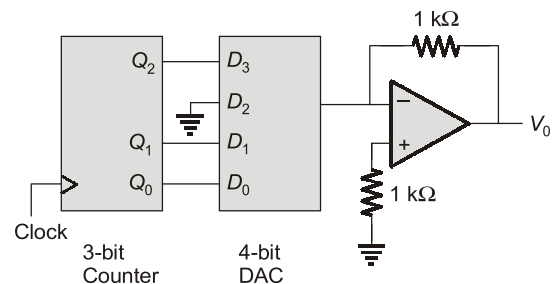


- (a) 10 V (b) 5 V
(c) 4 V (d) 8 V [GATE-2000]

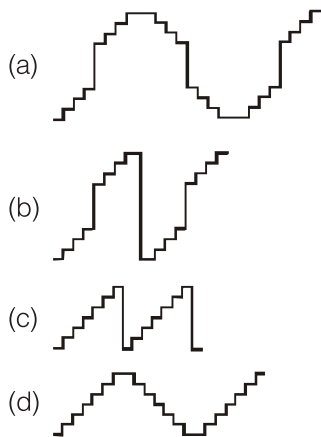
Q.5 The resolution of a 4-bit counting ADC is 0.5 Volts . For an analog input of 6.6 Volts , the digital output of the ADC will be

- (a) 1011 (b) 1101
(c) 1100 (d) 1110 [GATE-1999]

Q.6 A 4-bit D/A converter is connected to a free-running 3-bit UP counter, as shown in the following figure. Which of the following waveforms will be observed at V_0 ?



In the figure shown above, the ground has been shown by the symbol ⏏



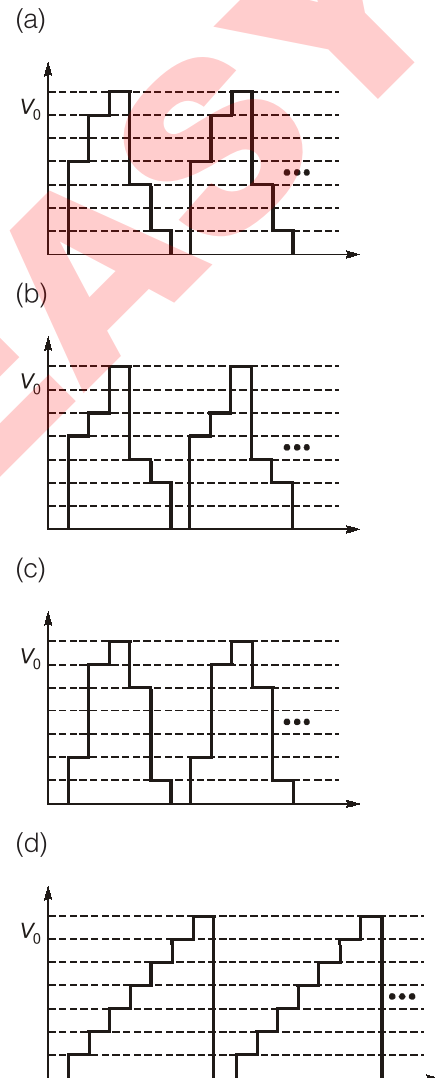
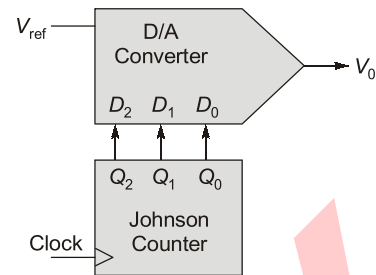
[GATE-2006]

23
69**Numerical Answer Type Questions**

- Q.7** An 8-bit D/A converter has a full scale output voltage of 20 V. The output voltage when the input is 11011011, is _____ V. [ESE-2001]
- Q.8** A 10-bit DAC provides an analog output which has a maximum value of 10.23 volts. Resolution of the DAC is _____ mV. [ESE-2012]
- Q.9** The analog output voltage of a 6 bit DAC with reference voltage as 20 V for the digital input 011101 is _____ Volts.
- Q.10** Consider a four bit D to A converter. The analog value corresponding to a digital signals of values 0000 and 0001 are 0 V and 0.0625 V respectively. The analog value (in Volts) corresponding to the digital signal 1111 is _____. [GATE-2015]

**Try Yourself**

- T1.** The output of a 3-stage Johnson (twisted-ring) counter is fed to a digital-to-analog (D/A) converter as shown in the figure below. Assume all states of the counter to be unset initially. The waveform which represents the D/A converter output V_0 is



[GATE-2011]

- T2.** Match **List-I** (Type of N -bit ADC) with **List-II** (Characteristics) and select the correct answer using the codes given below the lists:

List-I

- A. Flash converter
- B. Successive approximation
- C. Counter ramp
- D. Dual slope

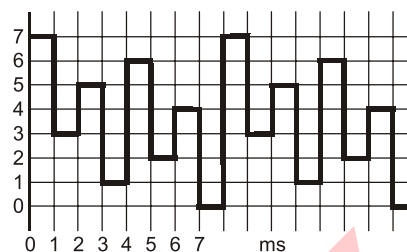
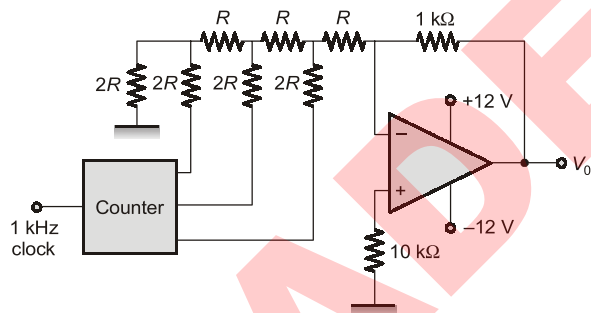
List-II

- 1. Integrating Type
- 2. Fastest converter
- 3. Maximum conversion time = N bits
- 4. Uses a DAC in its feedback path

Codes :

	A	B	C	D
(a)	2	3	4	1
(b)	1	3	4	2
(c)	2	4	3	1
(d)	1	4	3	2

- T3.** A student has made a 3-bit binary down counter and connected to the R - $2R$ ladder type DAC [Gain = $(-1 \text{ K}\Omega/2R)$] as shown in figure to generate a staircase waveform. The output achieved is different as shown in figure. What could be the possible cause of this error?



- (a) The resistance values are incorrect.
 - (b) The counter is not working properly.
 - (c) The connection from the counter to DAC is not proper.
 - (d) The R and $2R$ resistances are interchanged.
- T4.** The number of comparators in a 4-bit flash ADC is
- (a) 4
 - (b) 5
 - (c) 15
 - (d) 16

[GATE-2000]

■■■■

© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.



Semiconductor Memories



Multiple Choice Questions

- Q.1** Each cell of a static Random Access Memory contains
- 6 MOS transistors.
 - 4 MOS transistors and 2 capacitors
 - 2 MOS transistors and 4 capacitors
 - 1 MOS transistor and 1 capacitors

[GATE-1996]

- Q.2** In a DRAM,
- periodic refreshing is not required
 - information is stored in a capacitor
 - information is stored in a latch
 - both read and write operations can be performed simultaneously

[GATE-2017]

- Q.3** The minimum number of MOS transistors required to make a dynamic RAM cell is

- 1
- 2
- 3
- 4

[GATE-1995]

- Q.4** A dynamic RAM consists of

- 6 transistors
- 2 transistors and 2 capacitors
- 1 transistor and 1 capacitor
- 2 capacitors only

[GATE-1994]

- Q.5** Consider the following statements for a DRAM:

- Bit is stored as a charge.
- It is made of MOS transistors.
- Speed of DRAM is faster than processors.
- Each memory cell requires six transistors.

Which of these statements are correct?

- 1 and 2 only
- 2 and 3 only
- 3 and 4 only
- 1, 2, 3 and 4

[ESE-2012]

- Q.6** A ROM is to be used to implement a "squarer", which outputs the square of a 4-bit number. What must be the size of the ROM?

- 16 address lines and 16 data lines
- 4 address lines and 8 data lines
- 8 address lines and 8 data lines
- 4 address lines and 16 data lines

[ESE-2004]

- Q.7** A single ROM is used to design a combinational circuit described by a truth-table. What is the number of address lines in the ROM?

- Number of input variables in the truth table
- Number of output variables in the truth-table
- Number of input plus output variables in the truth-table
- Number of lines in the truth-table

[ESE-2006]

- Q.8** Which one of the following statements is correct?

- PROM contains a programmable 'AND' array and a fixed 'OR' array
- PLA contains a fixed 'AND' array and a programmable 'OR' array
- PROM contains a fixed 'AND' array and a programmable 'OR' array
- PLA contains a programmable 'AND' array and a programmable 'NOR' array

[ESE-2004]

© Copyright: Subject matter to MADE EASY Publications, New Delhi. No part of this book may be reproduced or utilised in any form without the written permission.

Q.9 Statement (I): PLA contains a fixed AND array and programmable OR array.

Statement (II): PROM contains a fixed AND array and a programmable OR array.

- (a) Both Statement I and Statement II are individually true and Statement II is the correct explanation of Statement I
- (b) Both Statement I and Statement II are individually true but Statement II is not the correct explanation of Statement I
- (c) Statement I is true but Statement II is false
- (d) Statement I is false but Statement II is true

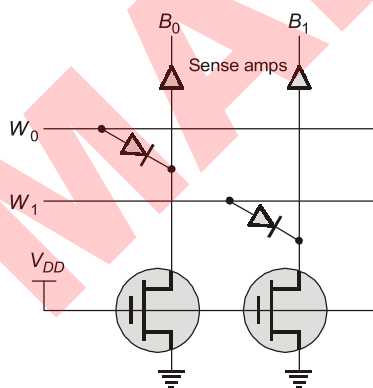
[ESE-2018]

Q.10 A PLA can be

- (a) as a microprocessor
- (b) as a dynamic memory
- (c) to realize a sequential logic
- (d) to realize a combinational logic.

[GATE-1994]

Q.11 A 2×2 ROM array is built with the help of diodes as shown in the circuit below. Here W_0 and W_1 are signals that select the word lines and B_0 and B_1 are signals that are output of the sense amps based on the stored data corresponding to the bit lines during the read operation.



$$\begin{matrix} B_0 & B_1 \\ W_0 [D_{00} & D_{01}] \\ W_1 [D_{10} & D_{11}] \end{matrix}$$

Bits stored in the ROM Array

During the read operation, the selected word line goes high and the other word line is in a high impedance state. As per the implementation shown in the circuit diagram above, what are the bits corresponding to D_{ij} (where $i = 0$ or 1 and $j = 0$ or 1) stored in the ROM?

- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- (b) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- (c) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

[GATE-2018]

23
69

Numerical Answer Type Questions

Q.12 A semiconductor RAM has a 12 bit address register and an 8 bit data register. The total number of bits in the memory is _____.

Q.13 A 16 kB (=16,384 bit) memory array is designed as a square with an aspect ratio of one (number of rows is equal to the number of columns). The minimum number of address lines needed for the row decoder is _____. [GATE-2015]

Q.14 The minimum number of MOS transistors required to make a dynamic RAM cell are _____.

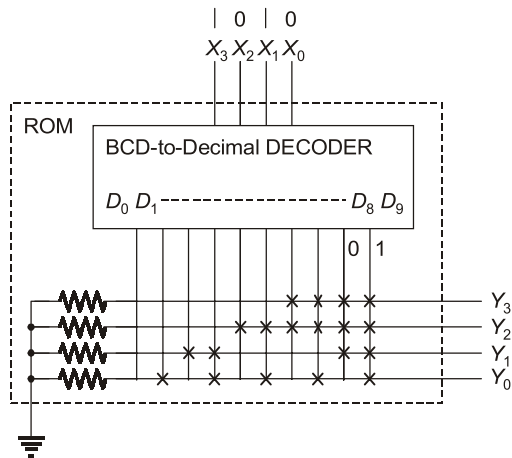
Q.15 The minimum number of MOS transistors required to make a static RAM cell are _____.

Q.16 It is desired to have 64×8 memory and if only 16×4 size chips are available then number of chips required are _____.



Try Yourself

- T1. If the input X_3, X_2, X_1, X_0 to the ROM in the figure are 8 4 2 1 BCD numbers, then the outputs Y_3, Y_2, Y_1, Y_0 are



- (a) gray code numbers
(b) 2 4 2 1 BCD numbers
(c) excess-3 code numbers
(d) none of the above

[2002 : 2 Marks]

- T2. If WL is the Word Line and BL the Bit Line, an SRAM cell is shown in

